



Project Title

A Thesis submitted for the degree of Master of Science in MSc Biomaterials and Tissue Engineering

**Comparison of Bubble Radius Dynamics in Newtonian and Viscoelastic Media
(Boundary Element Method for Calculating Acoustic Radiation Force and Torque Acting on Spherical Particles)**

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1 Abstract

We present a formulation of the Keller–Miksis equation for the nonlinear radial dynamics of ultrasound contrast-agent bubbles embedded in viscoelastic media. The model combines the compressible bubble dynamics of the Keller–Miksis formulation with the Marmottant lipid-monolayer model, accounting for shell buckling, elasticity, and rupture, and a fractional Voigt model to describe the surrounding viscoelastic medium. The fractional formulation introduces a memory-dependent response, extending the classical Voigt description to account for the influence of deformation history on the surrounding medium. The governing equations are formulated in terms of the nondimensional bubble radius and velocity, with the pressure difference across the bubble wall incorporating the contributions from gas pressure, surface tension, liquid viscosity, shell viscosity, acoustic excitation, and viscoelastic stresses. The formulation is validated against established bubble dynamics models and published results, including the Marmottant model and the Yang and Church formulation for viscoelastic media.

The uncoated fractional-Voigt medium reproduces the resonance-frequency and amplitude-damping trends reported in the literature for physiologically realistic tissue elasticities, and the coated shell model reproduces the Marmottant buckling and rupture transitions under matched driving conditions. In the final combined model, increasing the surrounding medium’s shear modulus from water ($G=0$) to liver ($G=40$ kPa) reduces the oscillation amplitude by approximately 21% at a fixed driving frequency, while the fractional order α has a comparatively weak effect on amplitude (a variation of order 1% between $\alpha = 1$ and $\alpha = 0.75$) under the conditions investigated, suggesting its influence is more readily expressed through transient decay and phase than through steady-state amplitude alone. Numerical convergence of the final model is confirmed to be well within tolerance at the time step used throughout this work.

The resulting framework provides a unified approach for examining the coupled influence of nonlinear shell mechanics and deformation-dependent viscoelasticity on ultrasound contrast-agent dynamics in tissue-like environments.

2 Introduction

Gas bubbles driven by acoustic fields are central to a range of medical, biomedical, and industrial applications, including ultrasound imaging, targeted

drug delivery, and sonochemistry [1]. Their behaviour under acoustic excitation is therefore important both for understanding cavitation phenomena and for controlling bubble-based technologies.

In diagnostic ultrasound, contrast agents enhance the distinction between neighbouring tissue types by introducing gas-filled microbubbles that strongly scatter the incident acoustic field [19, 4]. These bubbles are typically stabilised by lipid monolayers or polymer-based shells. The coating improves stability and can extend bubble lifetime in vivo [21, 23], but it also changes the mechanical response and can reduce oscillation amplitude. The resulting dynamics therefore depend on both the shell and the surrounding medium.

The mathematical description of bubble dynamics developed from classical incompressible collapse models, including the work of Besant and Rayleigh [2, 20], to formulations that account for driven oscillation and liquid compressibility. For the larger-amplitude acoustic excitation relevant to contrast agents, compressibility becomes important and motivates the use of models such as Keller–Miksis rather than the simpler Rayleigh–Plesset equation.

The present work focuses on two coupled features of a contrast agent. First, the surrounding medium is treated as viscoelastic rather than purely Newtonian, allowing tissue-like stiffness and dissipation to be represented. Second, the bubble is coated with a lipid monolayer whose surface tension varies with radius, allowing buckling, elastic stretching, and rupture to be represented [14]. The surrounding-medium formulation is developed from the linear Voigt model used by Yang and Church [24] and extended to a fractional Voigt form that incorporates deformation history. The constituent models are then checked against published results before being combined into a single formulation. This staged approach separates model development from model verification and provides clear limiting cases for assessing the final implementation.

3 Review

3.1 Free Bubble

A free, or uncoated, bubble has no stabilising shell and therefore provides a useful baseline for assessing the additional effects introduced by coating and by the surrounding medium. Although coatings can improve bubble stability, they can also modify and reduce the radial response, creating a balance between stability and acoustic activity [22]. The starting point for the present formulation is therefore the classical radial

equation for an uncoated bubble:

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 = \frac{P_L - P_\infty}{\rho_L} \quad (1)$$

Lamb [12] and Rayleigh [20] established the incompressible bubble-collapse framework, but this treatment is valid only for a cavity collapsing under a constant far-field pressure and cannot represent a driven, oscillating bubble under ultrasound. Plesset's vapour-bubble reformulation [18] and Noltingk and Neppiras's extension to a gas-filled bubble [17, 16] addressed this limitation, giving the viscous Rayleigh–Plesset equation used in numerical studies such as Lauterborn's [13] and given below.

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 = \frac{1}{\rho_L}(P_0 + \frac{2\sigma}{R_0} - P_v(\frac{R_0}{R})^{3\kappa} + P_v - \frac{2\sigma}{R} - \frac{4\mu_L\dot{R}}{R} - p_\infty(t)) \quad (2)$$

The Rayleigh–Plesset equation retains an important restriction, however: it assumes the surrounding liquid is incompressible, which breaks down once the bubble-wall velocity approaches the speed of sound in the medium – a regime routinely reached by ultrasound contrast agents driven near their resonance frequency (the coated-bubble validation in Section 8.4 uses a 2.9 MHz drive, for which wall velocities are non-negligible relative to c). Herring and Trilling addressed this by rederiving the equation to account for the energy stored in the compressible liquid during oscillation, but assumed a constant sound velocity in the surrounding medium, which limits the model's validity to moderate oscillation amplitudes.

Gilmore (1952) improved upon this assumption by employing the Kirkwood-Bethe hypothesis [8], which states that the wave velocity will be equal to the sum of the velocity of the fluid and the local velocity of sound at the bubble.

$$\left(1 - \frac{\dot{R}}{c}\right) R\ddot{R} + \frac{3}{2}\left(1 - \frac{\dot{R}}{3c}\right) \dot{R}^2 = \left(1 + \frac{\dot{R}}{c}\right) \frac{p_L - p_\infty(t)}{\rho} + \frac{R}{\rho c} \frac{dp_L}{dt} \quad (3)$$

Gilmore's Kirkwood–Bethe correction removes the constant-sound-velocity assumption but at the cost of a more involved formulation that is harder to combine cleanly with an arbitrary bubble-wall pressure difference. Keller and Miksis's formulation [11] retains first-order compressibility corrections while remaining written directly in terms of a pressure difference at the bubble wall, which is exactly the form needed to substitute

different shell and medium models (Section 6.3) without re-deriving the equation of motion each time. This is why the Keller–Miksis form, rather than the simpler Rayleigh–Plesset equation used in the volume-variation framework of Carreras-Casanova and Vanhille [5], is adopted as the base model throughout this work. We therefore use a form of this improved Keller–Miksis, where it is written in terms of pressure difference, and used within existing modeling [24];

$$\left(1 - \frac{\dot{R}}{c}\right) R\ddot{R} + \frac{3}{2}\left(1 - \frac{\dot{R}}{3c}\right) \dot{R}^2 = \left(1 + \frac{\dot{R}}{c}\right) \frac{p_a - p_I}{\rho} + \frac{R}{\rho c} \frac{d}{dt}[p_a - p_I] \quad (4)$$

with

$$P_a - P_I = p_g - \frac{2\sigma}{R} - P_0 + P_A \sin(\omega t) + 3 \int_R^\infty \frac{\tau_{rr}}{r} dr$$

where $R(t)$ is the instantaneous bubble radius, R_0 the equilibrium bubble radius, $\dot{R}$ the radial velocity of the bubble wall, and $\ddot{R}$ the radial acceleration of the bubble wall. The surrounding fluid is characterised by its density ρ and the speed of sound c in the medium.

The pressure terms are defined as follows: P_I = pressure immediately outside the bubble surface; $P_a(t)$ = pressure applied by the ultrasound (far-field pressure); P_g = gas pressure inside the bubble; P_0 = ambient (static) pressure; P_A = acoustic pressure amplitude.

3.2 Contrast Agent (Coated Bubble)

The first contrast agent model can be dated back to de Jong[10]. This modeling approach was similar to that of Fox and Herzfeld [9], but this contrast agent coating was modeled as a viscoelastic solid shell rather than a purely elastic outer layer.

A more rigorous derivation of this equation was carried out by Church [7] for a coated shell with a shell of finite thickness. However, this derivation neglects thermal and acoustic re-radiation - thus Morgan [15] used a modified version of Herring–Trilling to enable a more accurate investigation of radiation damping:

$$\begin{aligned} & \rho_L \left(R\ddot{R} + \frac{3}{2}\dot{R}^2 \right) \left(P_0 + \frac{2\sigma}{R} + \frac{2\chi_s}{R_0} \right) \\ & \left(1 - \frac{3\kappa\dot{R}}{c_L} \right) \left(\frac{R_0}{R} \right)^{3\kappa} - \frac{4\mu_L\dot{R}}{R} - \frac{2\sigma}{R} \left(1 - \frac{\dot{R}}{c_L} \right) \\ & - \frac{2\chi_s}{R} \left(1 - \frac{3\dot{R}}{c_L} \right) \left(\frac{R_0}{R} \right)^2 - \frac{12\mu_s d_s \dot{R}}{R(R-d_s)} - P_\infty(t) \end{aligned}$$

Morgan’s radiation-damping correction and Church’s finite-thickness shell [7] both treat the shell as a single continuous elastic (or viscoelastic) layer whose surface tension does not depend on the bubble’s instantaneous radius. This is a reasonable approximation for small-amplitude oscillations, but lipid monolayer shells used in practice are known to buckle under compression and rupture under sufficient tension – neither of which a constant or purely linear shell model can represent. Marmottant [14] addressed this directly by making the effective surface tension a piecewise function of radius, with distinct buckled, elastic, and ruptured regimes; this formulation is adopted here (Section 7) in preference to the earlier constant-shell models precisely because it captures the buckling and rupture transitions with experimental support for lipid-coated contrast agents [14, 6], which the present model needs in order to reproduce the shell-driven amplitude discontinuities validated in Section 8.4.3.

3.3 Surrounding Medium

For a physically representative model of an ultrasound contrast agent, the surrounding medium must also be considered. As the bubble travels through the body, it may interact with different tissue environments, ranging from softer biological tissue to stiffer regions. The surrounding medium can also influence cavitation behavior, making its mechanical properties important when modeling bubble dynamics.

Several viscoelastic descriptions are available for the surrounding medium, including Maxwell, Voigt, and fractional Voigt models. The Maxwell model places the elastic and viscous elements in series, whereas the Voigt model places them in parallel. For the present application, the Voigt model provides a useful baseline because it gives a bounded strain response and has already been applied to bubble dynamics in viscoelastic media by Yang and Church [24].

Its main limitation is that the classical Voigt model depends only on the instantaneous strain and strain rate and has no representation of deformation history. A fractional Voigt formulation replaces the integer-order viscous derivative with a Caputo derivative [3],

introducing a power-law memory kernel. The motivation for this extension is therefore specific: it adds dependence on prior deformation history while retaining the classical Voigt model as a limiting case. The present study uses this construction to assess whether the additional memory term changes the predicted bubble dynamics under the selected conditions.

4 Assumptions

The surrounding medium is assumed to be infinite and spherically symmetric about the bubble, so that the bubble radius $R(t)$ is spatially uniform and the problem reduces to a single radial coordinate. Modelling the medium as infinite, rather than as a finite tissue volume of realistic extent, is a standard simplification that avoids introducing boundary reflections into the problem and is adopted here for numerical tractability.

The gas filling the bubble is assumed to contribute negligible density, elasticity, and viscosity relative to the shell and surrounding fluid. This justifies neglecting the internal stress contribution from the gas phase in the stress integral of Eq. (1), so that only the external medium and shell need to be modelled explicitly.

A single, isolated bubble is considered throughout, following Stride [22]; bubble–bubble interactions are neglected, which is reasonable at the low volume fractions typical of a dilute contrast-agent suspension and substantially reduces the computational cost of the numerical implementation. At the diagnostic ultrasound frequencies considered (1–20 MHz), the resulting bubble-wall velocity (~ 10 m/s) remains much smaller than the speed of sound in either the shell or the surrounding fluid (~ 1500 m/s), so the density and viscosity of the medium are treated as constants rather than functions of the local strain rate.

An incompressible-medium assumption is also applied, giving the radial velocity field $u = (R^2/r^2)\dot{R}$ used throughout Section 5.

Table 1: Densities and viscosities of relevant fluids that we may use [22]

Medium	ρ (kg m ⁻³)	μ (Pa s)	G (kPa)
Water	1000	1.0×10^{-3}	0
Blood	1030	1.5×10^{-3}	43

with ρ and μ taken as constant for a given medium.

5 Formulation of Surrounding

Having identified the Fractional Voigt model as the most suitable representation of the surrounding

medium (Section 3.3), the governing equations for the linear Voigt model are first derived and nondimensionalised, before being extended to the fractional case.

5.1 Linear Voigt

The linear Voigt constitutive model, following Yang and Church [24], is given by

$$\tau_{rr} = 2(G\gamma_{rr} + \mu\dot{\gamma}_{rr}) \quad (5)$$

where γ_{rr} is the radial strain and $\dot{\gamma}_{rr} = \partial u / \partial r$ the radial strain rate, with u the radial velocity field and G the shear modulus. This stress is substituted into the pressure difference

$$P_a - P_I = p_g - \frac{2\sigma}{R} - P_0 + P_A \sin(\omega t) + 3 \int_R^\infty \frac{\tau_{rr}}{r} dr$$

to obtain the governing equation for an uncoated bubble in a linear viscoelastic medium.

5.1.1 Formulation of Linear Voigt

Before the constitutive relation can be substituted into the nondimensional Keller–Miksis equation (Section 6), the strain and strain-rate fields must first be expressed in terms of the bubble radius $R(t)$. The following subsections derive this relationship, beginning with the velocity field induced in the surrounding medium by the oscillating bubble wall.

5.1.2 Kinematics of the Surrounding Medium - Radial Velocity Field

For the assumed spherical and incompressible motion, the radial velocity field determines both the strain and strain-rate fields required by the constitutive law.

Assuming the surrounding medium is incompressible, conservation of volumetric flow rate across any spherical surface concentric with the bubble relates the radial velocity field to the bubble-wall velocity (Appendix A.1), giving the radial velocity field:

$$v_r(r, t) = \frac{R^2 \dot{R}}{r^2}$$

5.1.3 Strain rate tensor

Differentiating the velocity field with respect to r gives the radial strain rate:

$$\dot{\gamma}_{rr} = \frac{\partial v_r}{\partial r} = -\frac{2R^2 \dot{R}}{r^3}$$

The corresponding increase in circumference as the bubble expands gives rise to a tangential strain rate:

$$\dot{\gamma}_{\theta\theta} = \dot{\gamma}_{\phi\phi} = \frac{v_r}{r} = \frac{R^2 \dot{R}}{r^3}$$

Summing the three principal strain rates gives $\dot{\epsilon}_{rr} + \dot{\epsilon}_{\theta\theta} + \dot{\epsilon}_{\phi\phi} = 0$, confirming that the assumed incompressible velocity field produces zero volumetric strain rate, as required.

Since the Kelvin–Voigt constitutive relation requires both strain and strain rate, $\sigma = E\epsilon + \eta\dot{\epsilon}$, the strain field is derived next.

5.1.4 Strain Tensor

Integrating the strain-rate field with respect to time (Appendix A.2) gives the radial strain,

$$\gamma_{rr} = -\frac{2}{3r^3} (R^3 - R_0^3),$$

and substituting into the Kelvin–Voigt relation $\tau_{rr} = G\gamma_{rr} + \mu\dot{\gamma}_{rr}$ gives the radial stress:

$$\tau_{rr} = -\frac{2G}{3r^3} (R^3 - R_0^3) - \frac{2\mu R^2 \dot{R}}{r^3}$$

5.1.5 Kelvin-Voigt Stress Tensor

In general tensorial form, the radial and tangential Kelvin–Voigt stresses are

$$\tau_{rr} = 2(G\gamma_{rr} + \mu\dot{\gamma}_{rr})$$

and

$$\tau_{\theta\theta} = 2(G\gamma_{\theta\theta} + \mu\dot{\gamma}_{\theta\theta})$$

5.1.6 Momentum Balance

The constitutive equation describes the relation between stress and strain within the viscoelastic medium. The Keller–Miksis equation, however, requires the pressure contribution arising from these stresses at the bubble wall rather than the stress field itself. To relate the two, conservation of linear momentum is applied to the surrounding medium, yielding a stress integral that represents the additional pressure exerted by the viscoelastic material on the oscillating bubble (Appendix A.2):

$$\tau_{rr} - \tau_{\theta\theta} = -\frac{2G(R^3 - R_0^3)}{r^3} - \frac{6\mu R^2 \dot{R}}{r^3}$$

The only remaining unknown is therefore $\partial\sigma_{rr}/\partial r$, which is obtained from the radial momentum balance below.

5.1.7 Continuum Mechanics

The constitutive relation defines the stress within the surrounding viscoelastic medium; however, the contribution of this stress to the bubble dynamics must be obtained through the radial momentum balance. For an incompressible, spherically symmetric medium, the radial momentum equation is given by

$$\frac{\partial \tau_{rr}}{\partial r} + \frac{2}{r}(\tau_{rr} - \tau_{\theta\theta}) = -\frac{\partial p}{\partial r}.$$

Integrating this expression using the linear Voigt constitutive relation (Eq. 4) together with the strain (Eq. 5) and strain-rate (Eq. 6) fields derived above (Appendix A.3) gives the viscoelastic contribution to the pressure difference:

$$3 \int_R^\infty \frac{\tau_{rr}}{r} dr = -\frac{4G}{3} \left(1 - \frac{R_0^3}{R^3}\right) - 4\mu \frac{\dot{R}}{R} \quad (6)$$

The first term represents the elastic contribution associated with the shear modulus G , while the second represents viscous resistance through the surrounding-medium viscosity μ . This expression is consistent with the linear Voigt formulation used by Yang and Church [24] for modeling a bubble in a viscoelastic medium.

5.1.8 Non-dimensionalisation of the Linear Voigt Model

To obtain a dimensionless form, the same characteristic length (R_0) and time ($T = 1/f$) scales used to nondimensionalise the Keller–Miksis equation (Section 6.1) are applied here (Appendix A.3). Applying the pressure scaling $T^2/(\rho R_0^2)$ gives the dimensionless linear Voigt stress contribution,

$$\left(3 \int_R^\infty \frac{\tau_{rr}}{r} dr\right)_{DL} = -\frac{4GT^2}{3\rho R_0^2} \left(1 - \frac{1}{x_1^3}\right) - \frac{4\mu T}{\rho R_0^2} \frac{x_2}{x_1} \quad (7)$$

5.2 Fractional Voigt Extension

To extend the linear Voigt model to account for deformation memory, the integer-order time derivative in the viscous term is replaced by a Caputo fractional derivative of order α (Appendix A.4), giving the numerical representation of $D_t^\alpha (R^3 - R_0^3)$:

$$3 \int_R^\infty \frac{\tau_{rr}}{r} dr = -\frac{4G}{3} \left(1 - \frac{R^3}{R_0^3}\right) - \frac{4\eta_\alpha}{3R^3} D_t^\alpha (R^3 - R_0^3)$$

where $0 < \alpha < 1$ and D_t^α denotes the Caputo fractional derivative [3], defined as

$${}^c D_t^\alpha f(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t \frac{\dot{f}(s)}{(t-s)^\alpha} ds, 0 < \alpha < 1$$

giving the fractional Voigt stress relation

$$\tau_{rr} = 2[G\gamma_{rr} + \eta_\alpha {}^c D_t^\alpha \gamma_{rr}]$$

Because the Caputo derivative weights the entire strain-rate history rather than only the instantaneous strain rate, this formulation allows the medium's response to depend on its deformation history, consistent with the memory effects reported for biological tissue [3]. This motivates the extension from the linear Voigt model to the fractional Voigt model developed below.

5.2.1 Fractional Voigt consistency check

As a consistency check, we confirm that the fractional formulation

$$3 \int_R^\infty \frac{\tau_{rr}}{r} dr = -\frac{4G}{3} \left(1 - \frac{R^3}{R_0^3}\right) - \frac{4\eta_\alpha}{3R^3} D_t^\alpha (R^3 - R_0^3)$$

reduces correctly to the linear Voigt result when $\alpha = 1$. In this limit the Caputo derivative reduces to the standard first derivative,

$$D_t^1 (R^3 - R_0^3) = 3R^2 \dot{R},$$

so that

$$-\frac{4\eta_\alpha}{3R^3} D_t^1 (R^3 - R_0^3) = -4\eta_\alpha \frac{\dot{R}}{R}.$$

Identifying $\eta_\alpha = \mu$ then recovers the linear Voigt result,

$$-\frac{4G}{3} \left(1 - \frac{R^3}{R_0^3}\right) - 4\mu \frac{\dot{R}}{R},$$

confirming that the fractional formulation reduces correctly to its integer-order limit.

5.2.2 Fractional Voigt Extension of the Yang Model

The validated Yang formulation is extended by replacing the linear Voigt viscous contribution with the fractional Voigt constitutive formulation introduced in Section 5.2. The Keller–Miksis formulation, state variables and remaining pressure contributions are retained, while the surrounding-medium viscous contribution is replaced by a fractional memory term.

For $0 < \alpha < 1$, the Caputo fractional derivative is defined as

$${}^C D_t^\alpha f(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t \frac{\dot{f}(s)}{(t-s)^\alpha} ds.$$

Therefore, the dimensionless fractional viscoelastic stress contribution becomes (Appendix A.4):

$$\left(3 \int_R^\infty \frac{\tau_{rr}}{r} dr\right) = -\frac{4GT^2}{3\rho R_0^2} \left(1 - \frac{1}{x_1^3}\right) - \frac{4\eta_\alpha T^{2-\alpha}}{3\rho R_0^2 x_1^3} {}^C D_\tau^\alpha (x_1^3 - 1). \quad (8)$$

For convenience, the dimensionless fractional memory term is defined as

$$\mathcal{M}_\alpha(\tau) = {}^C D_\tau^\alpha (x_1^3 - 1). \quad (9)$$

5.2.3 Final Fractional Extension

The fractional extension is obtained by replacing the linear Voigt viscous contribution in the validated Yang pressure difference with the fractional memory term derived above. The remaining pressure contributions and the Keller–Miksis formulation are unchanged. Thus,

$$\Delta p_F = \frac{P_{g0} T^2}{\rho R_0^2} x_1^{-3\kappa} - \frac{2\sigma T^2}{\rho R_0^3 x_1} - \frac{P_0 T^2}{\rho R_0^2} + \frac{P_A T^2}{\rho R_0^2} \sin(2\pi\tau) - \frac{4GT^2}{3\rho R_0^2} \left(1 - \frac{1}{x_1^3}\right) - \frac{4\eta_\alpha T^{2-\alpha}}{3\rho R_0^2 x_1^3} \mathcal{M}_\alpha(\tau), \quad (10)$$

where

$$\mathcal{M}_\alpha(\tau) = {}^C D_\tau^\alpha (x_1^3 - 1). \quad (11)$$

The fractional formulation therefore retains the gas pressure, surface tension, ambient pressure, acoustic excitation, and elastic contributions of the Yang model. The linear viscous term is replaced by a fractional memory contribution governed by the fractional order α and coefficient η_α . Consequently, the present formulation reduces to the linear Voigt model in the appropriate integer-order limit.

6 Formulation of Keller–Miksis

Having identified the Fractional Voigt model as the most suitable representation of the surrounding medium (Section 3.3), we now formulate the governing bubble-dynamics equation. The Keller–Miksis formulation is adopted because it is written directly in terms of a pressure difference at the bubble wall, which allows different shell and medium models to

be substituted without re-deriving the equation of motion (Section 3.1). The Fractional Voigt model is used because previous work has demonstrated its applicability in the low-megahertz driving regime, for which comparative data are available [24], allowing the present formulation to be validated directly against previous results.

The gas pressure, p_g , is assumed to follow a polytropic process and is given by

$$p_{g0} \left(\frac{R_0}{R}\right)^{3\kappa}$$

where P_{g0} is the equilibrium gas pressure, R_0 the equilibrium radius, and κ the polytropic exponent. Substituting this into the pressure difference of Eq. (1) gives

$$P_a - P_l = p_{g0} \left(\frac{R_0}{R}\right)^{3\kappa} - \frac{2\sigma}{R} - P_0 + P_A \sin(\omega t) + 3 \int_R^\infty \frac{\tau_{rr}}{r} dr$$

6.1 Nondimensionalisation of Second Order

The governing equation is a second-order ordinary differential equation and must be reformulated as a first-order coupled system for numerical implementation. Both the Keller–Miksis equation and the pressure difference are nondimensionalised using the same characteristic scales; the pressure difference itself, however, is left in dimensional form at this stage, since two different pressure differences (Yang and Marmottant) are used to validate the model separately in Section 8.

The nondimensional parameters are built from three characteristic scales: the acoustic period T , the reciprocal of the driving frequency f ; the equilibrium bubble radius R_0 , in metres; and the atmospheric pressure P_0 , in pascals ($kg/(m \cdot s^2)$). Introducing the nondimensional time and radius variables $\tau = t/T$ and $x_1 = R/R_0$ (Appendix A.5) and substituting into the general Keller–Miksis equation gives

$$\begin{aligned} & \left(1 - \frac{R_0}{cT} \frac{dx_1}{d\tau}\right) R_0 x_1 \frac{R_0}{T^2} \frac{d^2 x_1}{d\tau^2} + \\ & \frac{3}{2} \left(1 - \frac{R_0}{3cT} \frac{dx_1}{d\tau}\right) \left(\frac{R_0}{T} \frac{dx_1}{d\tau}\right)^2 = \\ & \left(1 + \frac{R_0}{cT} \frac{dx_1}{d\tau}\right) \frac{\Delta P}{\rho} + \frac{R_0 x_1}{\rho cT} \frac{d\Delta P}{dt} \end{aligned} \quad (12)$$

Dividing through by $\frac{R_0^2}{T^2}$ gives

$$\left(1 - \frac{R_0}{cT}x_1'\right)x_1x_1'' + \frac{3}{2}\left(1 - \frac{R_0}{3cT}x_1'\right)(x_1')^2 = \left(1 + \frac{R_0}{cT}x_1'\right)\Delta p + \frac{Tx_1}{\rho c R_0} \frac{d\Delta P}{d\tau} \quad (13)$$

where

$$\Delta p = \frac{T^2}{\rho R_0^2} \Delta P, \quad \Delta P = \frac{\rho R_0^2}{T^2} \Delta p, \quad \frac{d\Delta P}{dt} = \frac{\rho R_0^2}{T^3} \frac{d\Delta p}{d\tau} \quad \text{where the lower integration limit corresponds to the instantaneous bubble wall, } r = R = R_0x_1, \text{ and } \dot{R} = \frac{R_0}{T}x_2.$$

and primes, x_1' , denote differentiation with respect to τ . Δp is the dimensionless pressure difference and ΔP its dimensional counterpart; the specific form of each is introduced separately for the Yang and Marmottant models in Section 6.3.

6.2 Coupled First Order for Numerical Implementation

The nondimensionalised Keller-Miksis equation remains second order. To integrate it numerically, auxiliary state variables are introduced (Appendix A.5), and isolating $dx_2/d\tau$ gives the nondimensional first-order Keller-Miksis system:

$$\frac{dx_2}{d\tau} = \frac{\left(1 + \frac{R_0}{cT}x_2\right)\Delta p + \frac{R_0x_1}{cT} \frac{d\Delta p}{d\tau} - \frac{3}{2}\left(1 - \frac{R_0}{3cT}x_2\right)x_2^2}{\left(1 - \frac{R_0}{cT}x_2\right)x_1} \quad (14)$$

with

$$\Delta p = \frac{T^2}{\rho R_0^2} \Delta P \quad (15)$$

and

$$\Delta P = P_a - P_I \quad (16)$$

Here, $\Delta P = P_a - P_I$ denotes the dimensional pressure difference, while Δp denotes the corresponding dimensionless pressure difference obtained through the pressure scaling, $\frac{T^2}{\rho R_0^2}$. The pressure difference is retained in this general form so that model-specific formulations can be introduced subsequently.

It is important to distinguish x_1 , which represents the dimensionless instantaneous bubble radius, from r , which represents the spatial coordinate in the surrounding medium. Therefore, the dimensional instantaneous bubble radius is therefore related to the dimensionless radius by $R = R_0x_1$, while the radial coordinate r is retained as the integration variable

when evaluating the stress contribution. Consequently, the lower limit of the stress integral can be written as $r = R = R_0x_1$.

Therefore

$$3 \int_R^\infty \frac{\tau_{rr}}{r} dr$$

The Keller-Miksis equation is second order in the bubble radius. For numerical integration, it is reformulated as an equivalent system of coupled first-order ordinary differential equations by introducing the dimensionless bubble radius and radial velocity as state variables.

With the general equation now nondimensionalised and reduced to first order, the two model-specific pressure differences used to validate the bubble dynamics are introduced below.

6.3 Pressure Differences

Two published bubble-dynamics results are used to validate the model: Yang and Church [24], who model an uncoated bubble in a linear viscoelastic medium, and Marmottant [14], who model a coated bubble in a Newtonian medium (water). Both are recovered from the general Keller-Miksis system derived above by substituting a different pressure difference ΔP ; once each is validated independently (Section 8), the medium formulation from the first and the shell formulation from the second are combined into the final model (Section 9).

6.3.1 Pressure difference - Yang

For an uncoated bubble, the pressure difference is

$$\Delta P(Uncoated) = \left[P_g - \frac{2\sigma}{R} - P_0 + P_A \sin(\omega t) + 3 \int_R^\infty \frac{\tau_{rr}}{r} dr \right] \quad (17)$$

with

$$3 \int_R^\infty \frac{\tau_{rr}}{r} dr = -\frac{4G}{3} \left(1 - \frac{R_0^3}{R^3}\right) - 4\mu \frac{\dot{R}}{R}$$

giving

$$\Delta P(\text{Uncoated}) = P_g - \frac{2\sigma}{R} - P_0 + P_A \sin(\omega t) - \frac{4G}{3} \left(1 - \frac{R_0^3}{R^3}\right) - 4\mu \frac{\dot{R}}{R} \quad (18)$$

where G is the elastic modulus and μ is the viscosity parameter. Substituting the gas pressure,

$$p_g = p_{g0} \left(\frac{R_0}{R}\right)^{3\kappa}$$

gives

$$\Delta P(\text{Uncoated}) = p_{g0} \left(\frac{R_0}{R}\right)^{3\kappa} - \frac{2\sigma}{R} - P_0 + P_A \sin(\omega t) - \frac{4G}{3} \left(1 - \frac{R_0^3}{R^3}\right) - 4\mu \frac{\dot{R}}{R} \quad (19)$$

Introducing the nondimensional variables of Section 6.1 (Appendix A.6), this formulation provides the general first-order form of the Keller–Miksis equation. The pressure difference ΔP is retained as a model-dependent term and is defined separately for the Yang and Marmottant formulations.

Substituting $\Delta p = \Delta p_Y$ gives

$$\Delta p_Y = \frac{P_{g0} T^2}{\rho R_0^2} x_1^{-3\kappa} - \frac{2\sigma T^2}{\rho R_0^3 x_1} - \frac{P_0 T^2}{\rho R_0^2} + \frac{P_A T^2}{\rho R_0^2} \sin(2\pi\tau) - \frac{4GT^2}{3\rho R_0^2} \left(1 - \frac{1}{x_1^3}\right) - \frac{4\mu T}{\rho R_0^2} \frac{x_2}{x_1} \quad (20)$$

6.3.2 Derivation of Yang

Since the dimensionless Yang pressure difference, Δp_Y , is a function of the dimensionless bubble radius x_1 , dimensionless radial velocity x_2 , and dimensionless time τ , its total derivative with respect to τ is obtained using the multivariable chain rule. Thus,

$$\frac{d\Delta p_Y}{d\tau} = \frac{\partial \Delta p_Y}{\partial x_1} \frac{dx_1}{d\tau} + \frac{\partial \Delta p_Y}{\partial x_2} \frac{dx_2}{d\tau} + \frac{\partial \Delta p_Y}{\partial \tau}. \quad (21)$$

Since the first state equation is

$$\frac{dx_1}{d\tau} = x_2, \quad (22)$$

the total derivative can be written as

$$\frac{d\Delta p_Y}{d\tau} = \frac{\partial \Delta p_Y}{\partial x_1} x_2 + \frac{\partial \Delta p_Y}{\partial x_2} \frac{dx_2}{d\tau} + \frac{\partial \Delta p_Y}{\partial \tau}. \quad (23)$$

This formulation accounts for changes in the pressure difference arising from variations in the

bubble radius, bubble-wall velocity, and the explicitly time-dependent acoustic excitation.

We derive with respect to x_1 then x_2 , and finally with respect to τ ;

$$\begin{aligned} \frac{d\Delta p_Y}{d\tau} = & \left[-\frac{3\kappa P_{g0} T^2}{\rho R_0^2} x_1^{-3\kappa-1} + \frac{2\sigma T^2}{\rho R_0^3 x_1^2} \right. \\ & \left. - \frac{4GT^2}{\rho R_0^2 x_1^4} + \frac{4\mu T}{\rho R_0^2} \frac{x_2}{x_1^2} \right] x_2 \\ & - \frac{4\mu T}{\rho R_0^2 x_1} \frac{dx_2}{d\tau} + \frac{2\pi P_A T^2}{\rho R_0^2} \cos(2\pi\tau) \end{aligned} \quad (24)$$

The total derivative of the dimensionless Yang pressure difference is then substituted into the general first-order Keller–Miksis equation. Because Δp_Y contains the bubble-wall velocity through the viscous term, its total derivative introduces $dx_2/d\tau$. This acceleration-dependent contribution is therefore collected with the other acceleration terms on the left-hand side before solving explicitly for $dx_2/d\tau$.

The final pressure difference and Keller–Miksis equation for the uncoated validation is

$$\frac{\partial \Delta p_Y}{\partial x_2} = -\frac{4\mu T}{\rho R_0^2 x_1}, \quad (25)$$

$$\begin{aligned} \left[\left(1 - \frac{R_0}{cT} x_2\right) x_1 + \frac{4\mu}{\rho c R_0} \right] \frac{dx_2}{d\tau} = & \left(1 + \frac{R_0}{cT} x_2\right) \Delta p_Y \\ & + \frac{R_0 x_1}{cT} \left[\frac{\partial \Delta p_Y}{\partial x_1} x_2 + \frac{\partial \Delta p_Y}{\partial \tau} \right] - \frac{3}{2} \left(1 - \frac{R_0}{3cT} x_2\right) x_2^2. \end{aligned} \quad (26)$$

then

$$\frac{dx_1}{d\tau} = x_2. \quad (27)$$

$$\frac{dx_2}{d\tau} = \frac{N_Y}{D_Y}. \quad (28)$$

where

$$\begin{aligned} N_Y = & \left(1 + \frac{R_0}{cT} x_2\right) \Delta p_Y + \frac{R_0 x_1}{cT} \left[\frac{\partial \Delta p_Y}{\partial x_1} x_2 + \frac{\partial \Delta p_Y}{\partial \tau} \right] \\ & - \frac{3}{2} \left(1 - \frac{R_0}{3cT} x_2\right) x_2^2 \end{aligned} \quad (29)$$

with

$$D_Y = \left(1 - \frac{R_0}{cT} x_2\right) x_1 + \frac{4\mu}{\rho c R_0}. \quad (30)$$

6.3.3 Pressure difference - MAR

The same nondimensionalisation and first-order formulation described for the Yang model are applied to the Marmottant formulation. The state variables x_1 and x_2 , and the first-order Keller–Miksis system, remain unchanged, with the model-specific behaviour incorporated entirely through the pressure difference. This pressure difference includes the Marmottant shell model, $\sigma(x_1)$, a radius-dependent surface tension (Section 7), together with a shell viscous term $\frac{4\kappa_s \dot{R}}{R^2}$ that has no counterpart in the Yang formulation above. The stress integral of Eq. (1) is neglected here, since this model assumes a Newtonian surrounding medium, specifically water.

Thus, with $\Delta p = \Delta p_M$,
with

$$\Delta p_M = \frac{P_{g0} T^2}{\rho R_0^2} x_1^{-3\kappa} - \frac{2\sigma(x_1) T^2}{\rho R_0^3 x_1} - \frac{P_0 T^2}{\rho R_0^2} + \frac{P_A T^2}{\rho R_0^2} \sin(2\pi\tau) - \frac{4\mu T}{\rho R_0^2} \frac{x_2}{x_1} - \frac{4\kappa_s T}{\rho R_0^3} \frac{x_2}{x_1^2} \quad (31)$$

Introducing the same dimensionless variables as before (Appendix A.7), the total pressure derivative follows the same procedure used for the Yang model (Appendix A.7). The acceleration-dependent term is again collected on the left-hand side of the first-order Keller–Miksis equation, giving

$$\begin{aligned} & \left[\left(1 - \frac{R_0}{cT} x_2 \right) x_1 \right. \\ & + \frac{4\mu}{\rho c R_0} + \frac{4\kappa_s}{\rho c R_0^2} \frac{dx_2}{d\tau} = \left(1 + \frac{R_0}{cT} x_2 \right) \Delta p_M \\ & \quad + \frac{R_0 x_1}{cT} \left[\frac{\partial \Delta p_M}{\partial x_1} x_2 + \frac{\partial \Delta p_M}{\partial \tau} \right] \\ & \quad \left. - \frac{3}{2} \left(1 - \frac{R_0}{3cT} x_2 \right) x_2^2. \right] \end{aligned} \quad (32)$$

so that

$$\frac{dx_2}{d\tau} = \frac{N_M}{D_M}. \quad (33)$$

where

$$\begin{aligned} N_M &= \left(1 + \frac{R_0}{cT} x_2 \right) \Delta p_M + \\ & \frac{R_0 x_1}{cT} \left[\frac{\partial \Delta p_M}{\partial x_1} x_2 + \frac{\partial \Delta p_M}{\partial \tau} \right] - \frac{3}{2} \left(1 - \frac{R_0}{3cT} x_2 \right) x_2^2. \end{aligned} \quad (34)$$

with

$$D_M = \left(1 - \frac{R_0}{cT} x_2 \right) x_1 + \frac{4\mu}{\rho c R_0} + \frac{4\kappa_s}{\rho c R_0^2} x_1. \quad (35)$$

7 Marmottant shell

The Marmottant model [14] allows the bubble to be coated in a lipid monolayer, incorporating both buckling and rupture behaviour, and can be integrated directly into the Keller–Miksis framework developed above by substituting its shell surface tension into the pressure difference of Section 6.3.2. The model captures three distinct regimes as the shell radius varies: a buckled state in which the shell carries no surface tension, an elastic compressible state, and a ruptured state in which the interface behaves as a clean gas–liquid boundary. The Marmottant theory replaces the constant surface tension σ used in Section 6.3.1 with a radius-dependent surface tension $\sigma(R)$, so that the shell’s viscous and elastic properties – and hence the final stress – depend explicitly on the instantaneous bubble radius. Rupture and buckling are included for a more realistic representation of lipid shell behaviour, and are determined by the radius of the microbubble as follows:

$$\sigma(R) = \begin{cases} 0, & R \leq R_b, \\ \chi \left(\frac{R^2}{R_b^2} - 1 \right), & R_b < R < R_r, \\ \sigma_w, & R \geq R_r. \end{cases} \quad (36)$$

where;

R = Instantaneous bubble radius, R_b = Buckling radius, below which the shell is buckled and the effective surface tension is zero, R_r = rupture radius, above which the shell is ruptured and the interface behaves as a clean gas liquid interface, χ = Shell elastic modulus, σ_w = Surface tension of the surrounding liquid (can assume to be water).

This monolayer lipid shell allows realistic behaviour of solid shell contrast agents to be captured. Buckling behaviour refers to the resistance solid shells exhibit under compression, which is released via bending of the shell rather than a further increase in surface tension [6]. Rupture, by contrast, occurs when the tension produced by shell stretching exceeds the shell’s structural strength [21].

7.1 Non-dimensional and first order

As with the other models, the rupture and buckling mechanics are presented in nondimensional form,

$$\sigma(x_1) = \begin{cases} 0, & R_0 x_1 \leq R_b, \\ \chi \left(\frac{R_0^2 x_1^2}{R_b^2} - 1 \right), & R_b < R_0 x_1 < R_r, \\ \sigma_w, & R_0 x_1 \geq R_r \end{cases} \quad (37)$$

with

$$R_r = R_b \sqrt{1 + \frac{\sigma_{breakup}}{\chi}}$$

where R_r is the rupture radius, needed to determine the bubble radius at which the shell will rupture.

8 Model Validation Against Literature

8.1 Numerical Implementation

The coupled first-order system arising from the Keller–Miksis equation (Eq. 14) was solved using two different numerical approaches, depending on whether the governing equations depend only on the instantaneous bubble state or additionally on its time history.

For the uncoated Newtonian bubble and the Marmottant-coated bubble in water, the governing equations depend only on the instantaneous state (x_1, x_2, τ) . These cases were integrated using SciPy’s `solve_ivp` with the implicit Radau method. The adaptive implicit scheme is appropriate for the rapid changes in bubble-wall acceleration that can occur during strong oscillation and near shell-regime transitions.

The fractional-Voigt models introduce a Caputo derivative whose value depends on the stored solution history. An adaptive solver would therefore require the memory term to be evaluated at trial states that are not yet part of the causal history. A fixed time-step RK4 scheme was consequently used, with the fractional memory term evaluated from the accumulated history on the outer time grid and held fixed during the four RK4 stages of each step. The RK4 scheme controls the local state update, while the accuracy of the complete coupled method is ultimately limited by the discretisation of the fractional memory term. This distinction is examined explicitly through the convergence study in Section 9.2. Early tests with explicit Euler produced systematic growth in oscillation amplitude over long integrations, so the higher-order RK4 scheme was adopted for the final calculations.

8.2 Linear Voigt - Yang

The final Keller–Miksis equation is first validated using the uncoated pressure difference, $\Delta P(Uncoated)$.

Yang and Church [24] use a linear Voigt model for its simplicity, and their results are reproduced here to validate the present Keller–Miksis formulation before extending it. The linear Voigt model is subsequently replaced by the fractional Voigt model developed in Section 5.2, which retains stress and strain memory through the Caputo derivative and is intended to represent soft tissue more physically than a model with no deformation history. Together with the nondimensional, first-order Keller–Miksis system, the linear Kelvin–Voigt model used in the previous research is applied first.

We implement the Yang formulation using the linear Voigt pressure contribution derived in Section 5.1 and the general first-order Keller–Miksis system derived in Section 6.2, before replacing the linear viscous term with the fractional formulation to represent a nonlinear viscoelastic medium.

We begin with the general Keller–Miksis equation nondimensionalised earlier;

$$\Delta p_Y = \frac{T^2}{\rho R_0^2} \Delta P_Y$$

with

$$\frac{\partial \Delta p_Y}{\partial x_2} = -\frac{4\mu T}{\rho R_0^2 x_1},$$

and total derivative (derived in Section 6.3.1)

$$\begin{aligned} \frac{d\Delta p_Y}{d\tau} = & \left[-\frac{3\kappa P_{g0} T^2}{\rho R_0^2} x_1^{-3\kappa-1} + \frac{2\sigma T^2}{\rho R_0^3 x_1^2} - \frac{4GT^2}{\rho R_0^2 x_1^4} \right. \\ & \left. + \frac{4\mu T}{\rho R_0^2} \frac{x_2}{x_1^2} \right] x_2 - \frac{4\mu T}{\rho R_0^2 x_1} \frac{dx_2}{d\tau} + \frac{2\pi P_A T^2}{\rho R_0^2} \cos(2\pi\tau). \end{aligned} \quad (38)$$

giving the Keller–Miksis system of Eq. 14 with N_Y , D_Y as defined in Section 6.3.1 (Appendix A.6). The following parameters are used to reproduce the validation case of Yang and Church [24].

8.2.1 Parameters

$P_0 = 1.01 \times 10^5$ Pa, $\rho = 1000$ kg/m³, $c = 1500$ m/s, $\sigma = 0.0725$ N/m, $k = 1.4$ $G = 0$, with G also varied up to 1.5 MPa in line with Yang and Church’s own parameter sweep [24], and $\mu = 0.005$ (blood; 0.015 for tissue).

8.2.2 Results

Figure 1 reproduces the dimensionless-radius result of Yang and Church [24] (their Figure 1a).

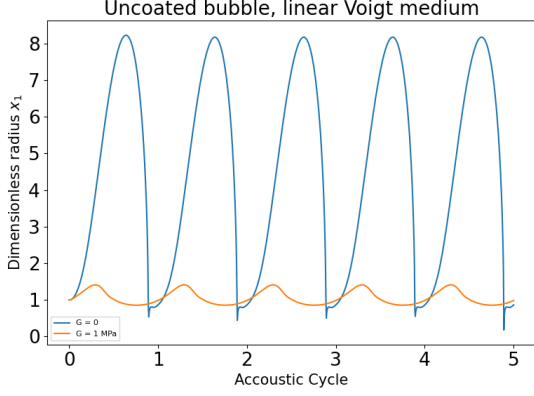


Figure 1: Dimensionless bubble radius x_1 for the uncoated linear-Voigt model, reproducing Yang and Church [24]. For $G = 0$ (red), the model gives $x_1 = 1.4$; for $G = 1$ MPa (blue), x_1 reaches upwards of 8

With the linear Voigt formulation now validated, the fractional model can be introduced and compared against these results and previous experimental findings.

8.3 Extension of Linear Model to Fractional

$$\Delta p_F = \frac{P_{g0}T^2}{\rho R_0^2} x_1^{-3\kappa} - \frac{2\sigma T^2}{\rho R_0^3 x_1} - \frac{P_0 T^2}{\rho R_0^2} + \frac{P_A T^2}{\rho R_0^2} \sin(2\pi\tau) - \frac{4GT^2}{3\rho R_0^2} \left(1 - \frac{1}{x_1^3}\right) - \frac{4\eta_\alpha T^{2-\alpha}}{3\rho R_0^2 x_1^3} \mathcal{M}_\alpha(\tau), \quad (39)$$

where

$$\mathcal{M}_\alpha(\tau) = {}^C D_\tau^\alpha (x_1^3 - 1). \quad (40)$$

Unlike the linear Voigt formulation, the fractional pressure difference contains a history-dependent memory term $\mathcal{M}_\alpha(\tau)$. Therefore, the total derivative is written in terms of the instantaneous radius and the fractional memory variable as

$$\frac{d\Delta p_F}{d\tau} = \frac{\partial \Delta p_F}{\partial x_1} x_2 + \frac{\partial \Delta p_F}{\partial \mathcal{M}_\alpha} \frac{d\mathcal{M}_\alpha}{d\tau} + \frac{\partial \Delta p_F}{\partial \tau}. \quad (41)$$

Since the Keller-Miksis equation requires $d\Delta p_F/d\tau$, the Caputo derivative definition is used to note that, for $f(\tau) = x_1^3 - 1$, the derivative of the memory term can be represented by the corresponding higher-order fractional derivative (Appendix A.8). Substituting the partial derivatives above into the same procedure used for the linear model gives (Appendix A.8)

with

$$\frac{dx_1}{d\tau} = x_2$$

$$\frac{dx_2}{d\tau} = \frac{\left(1 + \frac{R_0}{cT} x_2\right) \Delta p_F + \frac{R_0 x_1}{cT} \frac{d\Delta p_F}{d\tau} - \frac{3}{2} \left(1 - \frac{R_0}{3cT} x_2\right) x_2^2}{\left(1 - \frac{R_0}{cT} x_2\right) x_1}$$

Where every evaluation of $\mathcal{M}_\alpha(\tau)$ is obtained from the stored x_1 history using the L1 scheme.

8.3.1 Numerical Evaluation of the Fractional Derivative

Unlike the other validation cases, which use an adaptive Runge-Kutta solver (`solve_ivp`, Section 8.1), the fractional model requires a dedicated numerical scheme for the Caputo derivative, since fractional-order derivatives are non-local and cannot be evaluated from the instantaneous state alone. The fractional memory term is therefore evaluated on a uniform dimensionless time grid,

$$\tau_n = n\Delta\tau,$$

where $\Delta\tau$ is the dimensionless time step and n denotes the current time index.

The fractional memory quantity is defined as

$$\mathcal{M}_\alpha(\tau) = {}^C D_\tau^\alpha (x_1^3 - 1).$$

For $0 < \alpha < 1$, the Caputo derivative is approximated using the L1 discretisation. It replaces the function inside the fractional derivative with a piecewise linear interpolant.

Defining

$$f(\tau) = x_1^3(\tau) - 1,$$

the approximation at τ_n is

$$\mathcal{M}_\alpha^n \approx \frac{1}{\Gamma(2-\alpha)(\Delta\tau)^\alpha} \sum_{j=0}^{n-1} (f_{j+1} - f_j) [(n-j)^{1-\alpha} - (n-j-1)^{1-\alpha}] \quad (42)$$

Since

$$f_j = x_{1,j}^3 - 1,$$

this may be written as

$$\mathcal{M}_\alpha^n \approx \frac{1}{\Gamma(2-\alpha)(\Delta\tau)^\alpha} \sum_{j=0}^{n-1} (x_{1,j+1}^3 - x_{1,j}^3) [(n-j)^{1-\alpha} - (n-j-1)^{1-\alpha}] \quad (43)$$

The pressure derivative required by the Keller-Miksis equation contains the time derivative of the fractional memory quantity. For the present initial conditions,

$$x_1(0) = 1, \quad x_2(0) = 0,$$

the function $f(\tau) = x_1^3(\tau) - 1$ satisfies

$$f(0) = 0, \quad f'(0) = 0.$$

Consequently, the derivative of the Caputo memory term may be expressed as

$$\frac{d\mathcal{M}_\alpha}{d\tau} = {}^C D_\tau^{\alpha+1} (x_1^3 - 1).$$

For numerical implementation, the derivative of the computed memory history is evaluated using the causal backward difference,

$$\left(\frac{d\mathcal{M}_\alpha}{d\tau} \right)_n \approx \frac{\mathcal{M}_\alpha^n - \mathcal{M}_\alpha^{n-1}}{\Delta\tau}. \quad (44)$$

Substituting the memory term $\mathcal{M}_\alpha(\tau)$ into the pressure difference of Section 5.2.3 gives the fractional pressure difference used here (Appendix A.8), and the total derivative of the fractional pressure difference is therefore

$$\begin{aligned} \frac{d\Delta p_F}{d\tau} = & \left[-\frac{3\kappa P_{g0} T^2}{\rho R_0^2} x_1^{-3\kappa-1} + \frac{2\sigma T^2}{\rho R_0^3 x_1^2} - \right. \\ & \left. \frac{4GT^2}{\rho R_0^2 x_1^4} + \frac{4\eta_\alpha T^{2-\alpha}}{\rho R_0^2} \frac{\mathcal{M}_\alpha}{x_1^4} \right] x_2 \\ & - \frac{4\eta_\alpha T^{2-\alpha}}{3\rho R_0^2 x_1^3} \frac{d\mathcal{M}_\alpha}{d\tau} + \frac{2\pi P_A T^2}{\rho R_0^2} \cos(2\pi\tau). \end{aligned} \quad (45)$$

The fractional pressure difference is then substituted into the general first-order Keller-Miksis equation. Since the fractional memory term does not introduce an instantaneous dependence on x_2 , no additional $\partial\Delta p_F/\partial x_2$ acceleration term is introduced. Thus,

$$\frac{dx_1}{d\tau} = x_2$$

and

$$\frac{dx_2}{d\tau} = \frac{\left(1 + \frac{R_0}{cT} x_2\right) \Delta p_F + \frac{R_0 x_1}{cT} \frac{d\Delta p_F}{d\tau} - \frac{3}{2} \left(1 - \frac{R_0}{3cT} x_2\right) x_2^2}{\left(1 - \frac{R_0}{cT} x_2\right) x_1} \quad (46)$$

Introducing the fractional Voigt constitutive relation therefore introduces a history-dependent memory term through the Caputo fractional derivative [3]. Consequently, although the Keller-Miksis equation remains a differential governing equation, the resulting system is no longer a conventional memoryless ordinary differential equation in the instantaneous state variables

alone: the numerical solution requires retention of the full solution history in order to evaluate the fractional derivative at each time step, using the grid and L1 discretisation described in Section 8.3.1.

8.3.2 Parameters

Table 2: Parameters used for validation of the fractional Voigt model

Medium	ρ (kg m ⁻³)	μ (Pa s)	G (kPa)
Water	1000	1.8×10^{-3}	0
Liver	1100	9×10^{-3}	40

8.3.3 Results

The fractional Voigt model is validated using representative human tissue parameters, taking liver as the reference tissue [5].

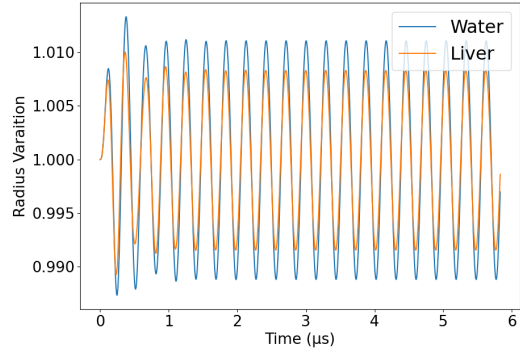


Figure 2: Dimensionless bubble radius x_1 for the uncoated fractional-Voigt model in water ($G = 0$) and liver ($G = 40$ kPa), $f = 3.43$ MHz, $p_0 = 5$ kPa.

Figure 2 compares the two media, where $G = 0$ represents the Newtonian properties of water and $G = 40$ kPa the shear properties of liver.

To further examine the effect of the fractional-Voigt formulation and the introduction of elasticity on bubble radius variation, the shear modulus is swept across $G = 3300$ (fat), 6700 (muscle), and 11000 Pa (glandular breast) [5], with all other parameters (density, viscosity) held at the values used for water.

To isolate the specific contribution of medium elasticity to bubble dynamics within the fractional-Voigt formulation, we conducted a controlled parametric sweep in which the shear modulus G was varied across a physiologically realistic range representative of soft biological tissues, while all other medium properties (density, viscosity, and fractional order α) were held

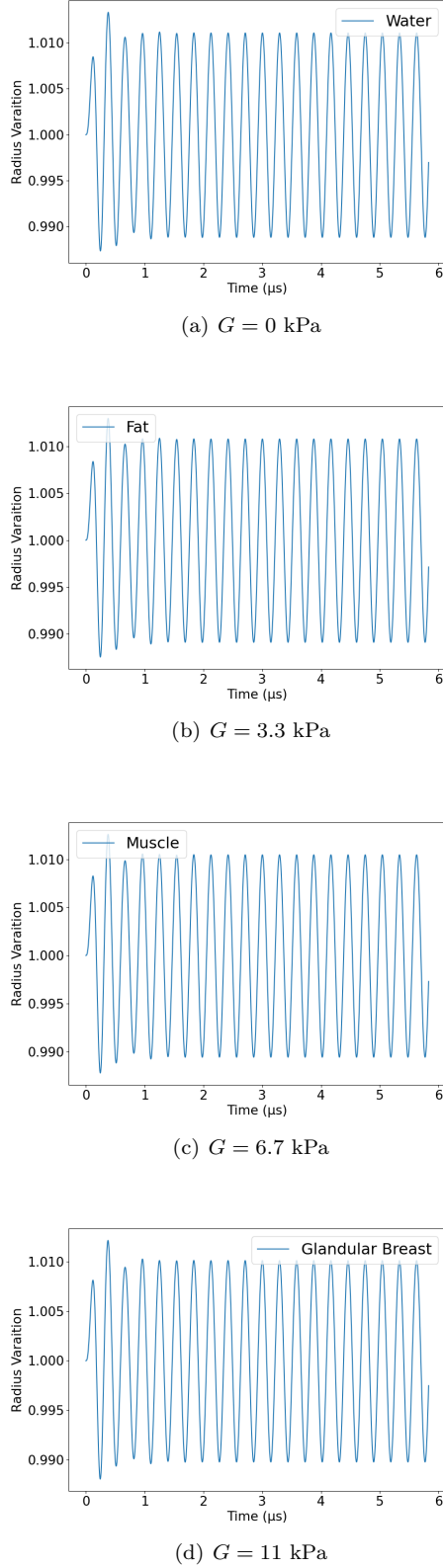


Figure 3: Bubble radius variation for the uncoated fractional-Voigt model at $f = 3.43$ MHz and $p_0 = 5$ kPa for increasing shear modulus G . All other parameters are held fixed.

Table 3: Shear moduli used in the soft-tissue parameter sweep.

Medium	G (kPa)
Water	0
Fat	3.3
Muscle	6.7
Glandular Breast	11

fixed at their water-baseline values. Across this range, the oscillation amplitude decreases only modestly with increasing shear modulus, consistent with the corresponding amplitude reduction in stiffer media reported by Carreras-Casanova and Vanhille [5] for the same physiologically realistic tissue set. Elasticity’s overall effect on the bubble response remains comparatively subtle relative to the substantially larger resonance-frequency changes reported in earlier studies employing shear moduli several orders of magnitude higher (0.5–1.5 MPa) [5]. This suggests that, within the range of shear elasticities relevant to soft biological tissue, elasticity exerts a real but comparatively subtle influence on stable-cavitation bubble dynamics relative to viscosity. This behavior is consistent with the underlying fractional-Voigt formulation, in which the elastic restoring term enters linearly and additively alongside the fractional viscous memory term, such that its relative contribution to the total restoring force scales directly with the magnitude of G relative to the medium’s other viscoelastic properties.

These trends are consistent with the amplitude-damping behaviour reported by Carreras-Casanova and Vanhille [5] for the same range of shear moduli, supporting the validity of the fractional-Voigt medium formulation.

8.4 Coated bubble within a Newtonian Fluid - Marmottant

With the shell model established (Section 7), the Marmottant pressure difference Δp_M derived in Section 6.3.2 is validated against a Newtonian surrounding medium (water), following its total derivative,

$$\begin{aligned}
 \frac{d\Delta p_M}{d\tau} = & \left[-\frac{3\kappa P_{g0} T^2}{\rho R_0^2} x_1^{-3\kappa-1} - \frac{2T^2}{\rho R_0^3} \frac{x_1 \sigma'(x_1) - \sigma(x_1)}{x_1^2} \right. \\
 & \left. + \frac{4\mu T}{\rho R_0^2} \frac{x_2}{x_1^2} + \frac{8\kappa_s T}{\rho R_0^3} \frac{x_2}{x_1^3} \right] x_2 \\
 & + \left[-\frac{4\mu T}{\rho R_0^2 x_1} - \frac{4\kappa_s T}{\rho R_0^3 x_1^2} \right] \frac{dx_2}{d\tau} + \\
 & \frac{2\pi P_A T^2}{\rho R_0^2} \cos(2\pi\tau).
 \end{aligned} \tag{47}$$

then

$$N_M = \left(1 + \frac{R_0}{cT}x_2\right) \Delta p_M + \frac{R_0 x_1}{cT} \left[\frac{\partial \Delta p_M}{\partial x_1} x_2 + \frac{\partial \Delta p_M}{\partial \tau} \right] - \frac{3}{2} \left(1 - \frac{R_0}{3cT}x_2\right) x_2^2 \quad (48)$$

with

$$D_M = \left(1 - \frac{R_0}{cT}x_2\right) x_1 + \frac{4\mu}{\rho c R_0} + \frac{4\kappa_s}{\rho c R_0^2 x_1} \quad (49)$$

$$\frac{dx_2}{d\tau} = \frac{N_M}{D_M} \quad (50)$$

$$\frac{dx_1}{d\tau} = x_2. \quad (51)$$

and

$$\sigma(x_1) = \begin{cases} 0, & R_0 x_1 \leq R_b, \\ \chi \left(\frac{R_0^2 x_1^2}{R_b^2} - 1 \right), & R_b < R_0 x_1 < R_r, \\ \sigma_w, & R_0 x_1 \geq R_r \end{cases} \quad (52)$$

with

$$R_r = R_b \sqrt{1 + \frac{\sigma_{breakup}}{\chi}}$$

With the partial derivatives;

$$\begin{aligned} \frac{\partial \Delta p_M}{\partial x_1} = & -\frac{3kP_{g0}T^2}{\rho R_0^2} x_1^{-3\kappa-1} - \frac{2T^2}{\rho R_0^3} \frac{x_1 \sigma'(x_1) - \sigma(x_1)}{x_1^2} \\ & + \frac{4\mu T}{\rho R_0^2} \frac{x_2}{x_1^2} + \frac{8\kappa_s T}{\rho R_0^3} \frac{x_2}{x_1^3}, \end{aligned} \quad (53)$$

8.4.1 Parameters

We use the same parameters [14] in order to compare and validate our results.

Therefore;

The density is taken as $\rho = 10^3 \text{ kg/m}^3$, the speed of sound in water, $c = 1480 \text{ m/s}$, the viscosity $\mu = 0.01 \text{ Pa s}$ and the ambient pressure, which is represented through standard atmospheric pressure $P_0 = 101.325 \text{ kPa}$.

The gas parameter $\kappa = 1.095$ is the ratio of specific heats for SF_6 , which behaves adiabatically; this is

appropriate since the thermal diffusion length in the gas over a high-frequency acoustic period is small compared to the bubble radius.

The shell lipid parameters are as follows: the surface dilatational viscosity $\kappa_s = 7.2 \times 10^{-9}$, representing a generic bubble baseline, and the equilibrium radius $R_0 = 0.975 \mu\text{m}$. The buckling radius is set equal to the equilibrium radius, $R_{buckling} = R_0 = 0.975 \mu\text{m}$, reflecting the assumption that the bubble is in a tension-free state initially. The breakup shell tension $\sigma_{breakup} = 0.13 \text{ N/m}$ represents a shell that ruptures more readily than SonoVue, whose buckling shell tension is 1 N/m . The surface tension of the surrounding water is $\sigma_{water} = 0.073 \text{ N/m}$.

The acoustic driving term, $P_A \sin(\omega t)$, uses an acoustic pressure amplitude $P_A = 130 \text{ kPa}$ and a frequency $f = \omega/2\pi = 2.9 \text{ MHz}$.

Since this is a coated bubble, $\sigma(R)$ is not constant and must be modelled dynamically using Eq. (36).

8.4.2 Results

The piecewise shell law produces the expected buckling and rupture transitions under the selected conditions, and this implementation is then used in the combined shell/medium model.

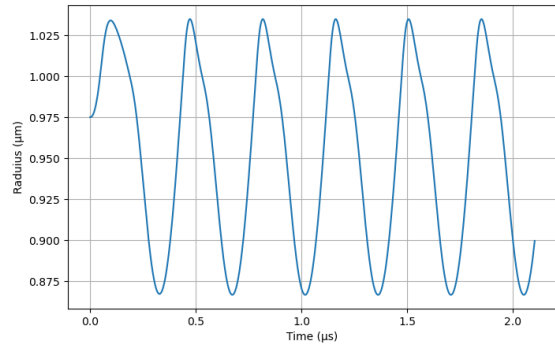


Figure 4: Marmottant coated-bubble radius in water, $f = 2.9 \text{ MHz}$, $P_A = 130 \text{ kPa}$, reproducing the validation case of [14].

A small difference in the maximum bubble oscillation peak is observed relative to [14]; this is attributed to the underlying bubble dynamics equation, since Marmottant's original model uses the Rayleigh–Plesset equation, whereas the present formulation uses the compressible Keller–Miksis equation throughout.

To further validate the shell model, the piecewise surface tension is tested specifically by tuning the driving parameters to induce shell rupture or buckling.

8.4.3 Rupture and Buckling

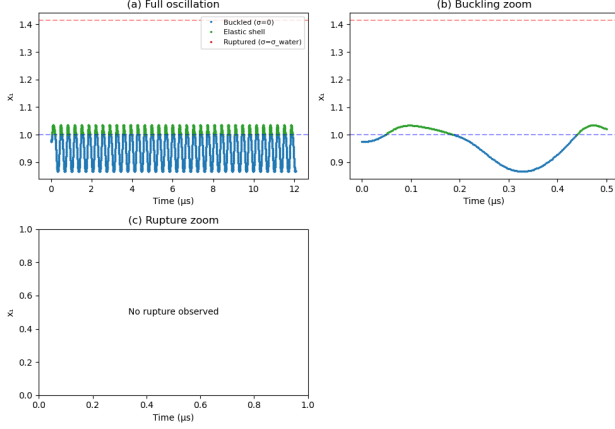


Figure 5: Marmottant model radius variation under the same conditions as Figure 4, shown at higher time resolution to illustrate the buckled and elastic regimes.

A rupture is then induced by increasing the driving acoustic pressure P_A or lowering the breakup tension $\sigma_{breakup}$, both of which lower the effective rupture threshold.

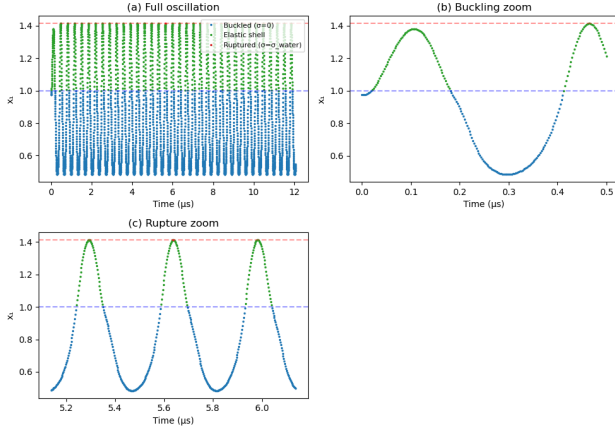


Figure 6: Marmottant coated-bubble radius with the driving acoustic pressure raised to $P_A = 1.2683$ MPa, sufficient to exceed the shell rupture threshold.

9 Final Model

Having validated the Keller–Miksis foundation, the fractional-Voigt medium model, and the Marmottant shell model independently (Section 8), the three are

now combined into a single formulation for a lipid-coated microbubble in a fractional-Voigt viscoelastic medium.

The Marmottant formulation (Section 7) retains a viscosity term describing a Newtonian surrounding fluid; this is replaced with the fractional Voigt model of Section 5.2, following the same linear-to-fractional extension used for the uncoated bubble in Section 8.3.

Combining the variables and pressure contributions from the constituent models gives the final fractional Voigt stress contribution

$$-\frac{4GT^2}{3\rho R_0^2} \left(1 - \frac{1}{x_1^3}\right) - \frac{4\eta_\alpha T^{2-\alpha}}{3\rho R_0^2 x_1^3} {}^C D_\tau^\alpha (x_1^3 - 1) \quad (54)$$

This is then substituted into the Marmottant pressure difference to account for a coated bubble in a fractional-Voigt medium, giving

$$\begin{aligned} \Delta p_M = & \frac{P_{g0}T^2}{\rho R_0^2} x_1^{-3\kappa} - \frac{2\sigma(x_1)T^2}{\rho R_0^3 x_1} - \frac{P_0T^2}{\rho R_0^2} \\ & + \frac{P_AT^2}{\rho R_0^2} \sin(2\pi\tau) - \frac{4GT^2}{3\rho R_0^2} \left(1 - \frac{1}{x_1^3}\right) - \\ & \frac{4\eta_\alpha T^{2-\alpha}}{3\rho R_0^2 x_1^3} {}^C D_\tau^\alpha (x_1^3 - 1) - \frac{4\kappa_s T}{\rho R_0^3} \frac{x_2}{x_1^2} \end{aligned} \quad (55)$$

where the total derivative becomes

$$\begin{aligned} \frac{d\Delta p_{\text{final}}}{d\tau} = & \left[-\frac{3\kappa P_{g0}T^2}{\rho R_0^2} x_1^{-3\kappa-1} - \frac{2T^2}{\rho R_0^3} \frac{x_1\sigma'(x_1) - \sigma(x_1)}{x_1^2} \right. \\ & - \frac{4GT^2}{\rho R_0^2 x_1^4} + \frac{4\eta_\alpha T^{2-\alpha}}{\rho R_0^2} \frac{\mathcal{M}_\alpha}{x_1^4} + \left. \frac{8\kappa_s T}{\rho R_0^3} \frac{x_2}{x_1^3} \right] x_2 \\ & - \frac{4\eta_\alpha T^{2-\alpha}}{3\rho R_0^2 x_1^3} \frac{d\mathcal{M}_\alpha}{d\tau} \\ & + \left[-\frac{4\kappa_s T}{\rho R_0^3 x_1^2} \right] \frac{dx_2}{d\tau} + \frac{2\pi P_AT^2}{\rho R_0^2} \cos(2\pi\tau). \end{aligned} \quad (56)$$

where σ' is

$$\sigma'(x_1) = \begin{cases} 0, & x_1 \leq \frac{R_b}{R_0}, \\ \frac{2\chi R_0^2}{R_b^2} x_1, & \frac{R_b}{R_0} < x_1 < \frac{R_r}{R_0}, \\ 0, & x_1 \geq \frac{R_r}{R_0}. \end{cases} \quad (57)$$

Substituting these into the general Keller–Miksis system, using the piecewise shell surface tension of Section 7, gives the final governing equations,

$$\frac{dx_1}{d\tau} = x_2, \quad (58)$$

where

$$\begin{aligned} \text{Numerator} = & \left(1 + \frac{R_0}{cT}x_2\right) \Delta p_{\text{final}} \\ & + \frac{R_0 x_1}{cT} \left[\frac{\partial \Delta p_{\text{final}}}{\partial x_1} x_2 \right. \\ & \left. + \frac{\partial \Delta p_{\text{final}}}{\partial \tau} + \frac{\partial \Delta p_{\text{final}}}{\partial \mathcal{M}_\alpha} \frac{d\mathcal{M}_\alpha}{d\tau} \right. \\ & \left. - \frac{3}{2} \left(1 - \frac{R_0}{3cT}x_2\right) x_2^2 \right] \end{aligned} \quad (59)$$

$$- \frac{3}{2} \left(1 - \frac{R_0}{3cT}x_2\right) x_2^2 \quad (60)$$

and

$$\text{Denominator} = \left(1 - \frac{R_0}{cT}x_2\right) x_1 + \frac{4\kappa_s}{\rho c R_0^2 x_1}. \quad (61)$$

which

$$\frac{dx_2}{d\tau} = \frac{\text{Numerator}}{\text{Denominator}}. \quad (62)$$

This final formulation includes both the Marmottant radius-dependent surface tension, together with its rupture and buckling dynamics, and the fractional Voigt representation of the surrounding medium in place of the simple Newtonian viscosity term used in the original Marmottant model. The coupled model is then assessed through limiting-case recovery, numerical convergence, and parametric response.

9.1 Verification and Validation

Before interpreting the fully coupled response, the implementation is checked against known limiting cases. The shell contribution is first disabled to recover the uncoated Keller–Miksis response in water and the corresponding viscoelastic response in liver. These checks isolate the surrounding-medium formulation before the shell and fractional terms are activated together. The parameters below are used, together with a dynamic viscosity $\mu = 1.002 \text{ mPa s}$ for water;

Table 4: Parameters used for validation of the Final Model

Medium	$\rho \text{ (kg m}^{-3}\text{)}$	$\mu \text{ (Pa s)}$	$G \text{ (kPa)}$
Water	1000	1.8×10^{-3}	0
Liver	1100	9×10^{-3}	40

9.2 Numerical Convergence

Since the final model introduces both a shell-driven discontinuity in $\sigma(x_1)$ at R_b and a fractional memory term with a cost that scales with the number of stored history points, its numerical convergence was checked separately from the constituent models validated in Section 8. The time step $\Delta\tau$ was varied between 8×10^{-3} and 1×10^{-3} , with a solution at $\Delta\tau = 5 \times 10^{-4}$ used as the reference, for both the water ($G = 0$) and liver ($G = 40 \text{ kPa}$) media at $\alpha = 1$.

The maximum absolute error in x_1 over the simulated interval decreases monotonically with decreasing $\Delta\tau$ for both media, falling from approximately $5\text{--}6 \times 10^{-4}$ at $\Delta\tau = 8 \times 10^{-3}$ to approximately 3×10^{-5} at $\Delta\tau = 1 \times 10^{-3}$ – more than two orders of magnitude smaller than the smallest oscillation amplitude reported in this work ($\approx 5 \times 10^{-3}$ for the α -sweep in Section 9.2). The observed order of convergence, estimated from successive halvings of $\Delta\tau$, is approximately 1.3–1.6, below the fourth-order accuracy of the underlying RK4 state update. This is attributed to the L1 discretisation of the Caputo fractional derivative (Section 8.3.1), whose local truncation error is $O(\Delta\tau^{2-\alpha})$ and therefore dominates the global error of the coupled scheme even though the state variables themselves are advanced with RK4. The time step $\Delta\tau = 1 \times 10^{-3}$ used throughout this section lies well within the converged regime for both media considered.

9.3 Results

Following the parameters in Table 4, the model foundation is validated with the Marmottant shell terms disabled (shell viscosity $\kappa_s = 0$ and the piecewise surface tension of Eq. (36) collapsed to the constant surface tension of water, so that no buckling or rupture regime is active), first using the water medium parameters ($G = 0$) to confirm that the model recovers the classical Keller–Miksis equation, and then using the liver medium parameters ($G = 40 \text{ kPa}$) to confirm that the surrounding-medium (fractional-Voigt) formulation reduces correctly to the expected Kelvin–Voigt behaviour once elasticity is reintroduced. A fixed driving frequency $f = 3.43 \text{ MHz}$ is used for both media, corresponding to one of the tissue-specific resonance frequencies at $R_0 = 1 \mu\text{m}$ identified by Carreras-Casanova and Vanhille [5]. This differs from the per-medium resonance driving used in the amplitude comparison of [5] (their Figure 6), which isolates the maximum achievable response in each medium; the fixed-frequency approach used here instead reflects a single clinical imaging frequency applied across tissue types, consistent with standard diagnostic ultrasound practice, and is sufficient for the present purpose of validating the model’s reduction to known limiting cases

rather than characterising each medium's resonance-optimised response.

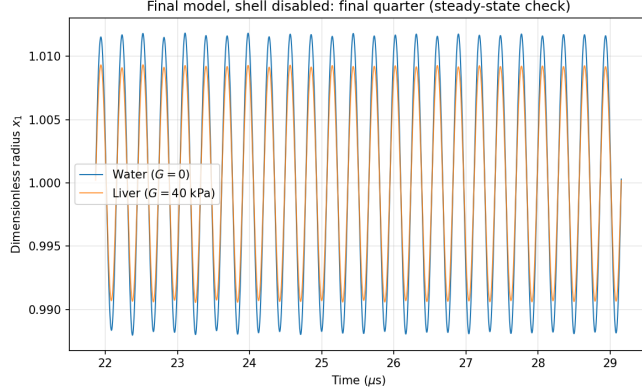
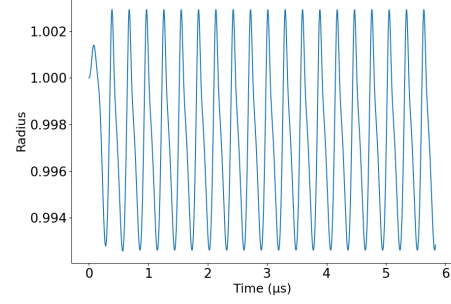


Figure 7: Final combined model with the Marmottant shell disabled, comparing the water ($G = 0$) and liver ($G = 40$ kPa) medium parameters of Table 4, $\alpha = 1$.

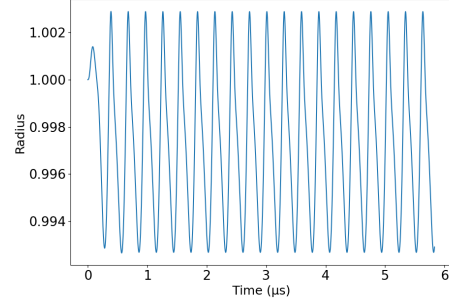
Figure 7 shows a smaller radial oscillation in liver than in water at the fixed driving frequency. Over the final quarter of the simulated interval, the measured amplitude is approximately 21% lower in the liver case. This is consistent with the additional elastic restoring contribution associated with the larger shear modulus. Because the two media are driven at the same frequency rather than at their individual resonances, however, the result should not be interpreted as a comparison of their maximum responses. The oscillation envelope also continues to decay over the simulated interval, so the quoted 21% difference is a finite-window measure rather than a fully steady-state quantity.

With this foundation confirmed, the shell terms are re-enabled and the effect of the fractional order α on the coupled bubble dynamics is examined, using the liver medium parameters with $G = 40$ kPa. The fractional order is varied between $\alpha = 1$ (the classical Voigt limit) and $\alpha = 0.75, 0.5$ (fractional-order responses).

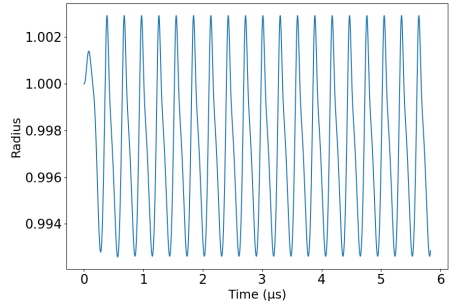
The resulting radius variations are shown in Figure 8. The three cases exhibit closely similar periodic responses, indicating that the oscillation amplitude is only weakly dependent on the fractional order for the conditions considered. For $\alpha = 1$, the minimum and maximum dimensionless radii are 0.99274 and 1.00275, respectively, giving an oscillation amplitude of 0.00501. Reducing the fractional order to $\alpha = 0.75$ gives an amplitude of 0.00495, a reduction of approximately 1.2%. At $\alpha = 0.5$, the amplitude returns to 0.00501, effectively matching the $\alpha = 1$ response. Thus, within the



(a) $\alpha = 1$



(b) $\alpha = 0.75$



(c) $\alpha = 0.5$

Figure 8: Dimensionless bubble radius x_1 for the final combined model in liver ($G = 40$ kPa) for fractional order $\alpha = 1, 0.75$, and 0.5 . All other parameters held fixed.

parameter range investigated, changing the fractional order produces only a small variation in radial oscillation amplitude. This suggests that the influence of the fractional memory term is not strongly expressed through amplitude alone under the present excitation conditions, and may instead be more apparent in other characteristics of the response, such as transient decay or phase.

10 Limitations

The surrounding medium is represented by a linear elastic contribution combined with a fractional viscous term. Real soft tissue can exhibit nonlinear stress-strain behaviour, particularly at larger strains. The present formulation should therefore be regarded as a moderate-deformation model, with accuracy expected to decrease as the bubble excursion moves further into the nonlinear tissue regime.

Thermal diffusion is not included. This is a standard simplification for the high-frequency, short-timescale oscillations considered here, but it means the model cannot capture thermal damping or heat-transfer-driven mass exchange, both of which become relevant at larger driving amplitudes or lower frequencies than those used in this work.

The model assumes the bubble remains spherically symmetric throughout its oscillation, so that the dynamics are described solely by the time-dependent radius $R(t)$. Although shell buckling and rupture are incorporated through the radius-dependent surface tension formulation, the model does not resolve non-spherical deformations, localised buckling patterns, or irregular shell geometries; it therefore captures the influence of buckling and rupture on the radial dynamics without accounting for associated shape instabilities or higher-order shape modes.

The fractional coefficient η_α is assigned the linear viscosity value μ so that the model recovers the classical Voigt form at $\alpha = 1$ (Section 5.2.1). This establishes a mathematical consistency condition, but it does not provide an experimental calibration of η_α for $\alpha < 1$. The fractional-order results should therefore be interpreted as a controlled model study rather than as quantitatively calibrated predictions of tissue rheology.

Finally, the foundation checks and water–liver comparison of Section 9.2 use a single fixed driving frequency across both media rather than driving each medium at its own resonance frequency, as discussed in that section; this is appropriate for the validation purpose intended there, but means the reported amplitude differences should not be interpreted as each medium’s maximum achievable response.

11 Discussion

The model development follows a staged verification strategy. The uncoated Keller–Miksis formulation is first reproduced using the Yang and Church framework [24]; the fractional-Voigt medium is then assessed against the stiffness-dependent trends reported by Carreras-Casanova and Vanhille [5]; and the Marmottant shell is checked against its reported buckling and rupture behaviour [14]. This sequence separates the governing bubble equation, the constitutive model, and the shell law before they are coupled.

The fully coupled formulation does not have a directly comparable published benchmark. It is therefore important to distinguish external validation from verification through known limiting cases. With the shell disabled, the formulation recovers the expected uncoated response; at $\alpha = 1$ with $\eta_\alpha = \mu$, the fractional formulation recovers the classical Voigt contribution. The convergence study further shows a systematic reduction in error as the time step is refined. These checks do not establish the accuracy of the fully coupled model in regimes for which no benchmark exists, but they provide evidence that the coupling preserves the behaviour of the constituent models.

The final results show that the response is more sensitive to the selected medium stiffness than to fractional order under the conditions studied. Increasing the shear modulus from water to liver reduces the measured radial amplitude, whereas changing α between 1 and 0.5 produces only a small change in amplitude. This does not imply that the fractional term is inactive: its effect may be more visible in phase or transient decay than in amplitude. The present results therefore support a restrained conclusion that, under the chosen forcing and parameterisation, stiffness produces the clearer change in radial amplitude.

A stronger test of the fractional contribution would use a resonance-matched frequency sweep and compare both phase and transient decay. Such a study would also separate changes in resonance location from changes in damping and restoring force, providing a more discriminating test of the physical role of the fractional term.

11.1 Wider context

Ultrasound contrast-agent modelling has relevance to safety and engineering design because shell rupture and large-amplitude oscillation are associated with stronger mechanical interactions with the surrounding

medium. The present framework provides a way to examine how these responses change when shell and medium properties are altered. However, the simplified assumptions and the lack of direct experimental validation mean that the model should not be used alone to define clinical exposure limits. Its more defensible role here is as a mechanistic tool for comparing the influence of shell and medium parameters.

The same modelling structure could also be adapted to other viscoelastic media, including polymeric or colloidal process fluids. Such applications would require independent characterisation of the relevant rheological parameters before quantitative predictions could be made.

11.2 Future Research

The main next step is to calibrate the fractional coefficient from experimental rheological data rather than identifying it directly with the linear viscosity. A resonance-matched, phase-resolved comparison across several values of α would then test whether fractional order produces measurable changes in resonance frequency, transient decay, or phase even when the amplitude response is small. At larger deformation amplitudes, a nonlinear or Zener-type constitutive law would provide a route beyond the present linear fractional-Voigt approximation. Finally, the coupled model could be used as a basis for reduced-order or surrogate models once the underlying parameter space has been more thoroughly characterised.

12 Conclusions

This work developed a nondimensional Keller–Miksis framework for an ultrasound contrast-agent bubble with a deformable lipid shell and a viscoelastic surrounding medium. The formulation separates the bubble dynamics from the model-specific wall pressure difference, allowing the shell and surrounding-medium descriptions to be combined within a common governing equation.

The surrounding medium was formulated first using a linear Voigt description and then extended with a Caputo fractional derivative to introduce deformation-history dependence. The Marmottant shell model was incorporated through a radius-dependent surface tension representing buckling, elastic stretching, and rupture. The constituent formulations were assessed against published bubble-dynamics behaviour before the coupled model was constructed.

For the final model, disabling the shell recovered the expected uncoated limiting response. In the fixed-frequency water–liver comparison, increasing the shear

modulus from $G = 0$ to 40 kPa reduced the measured oscillation amplitude by approximately 21% over the final quarter of the simulation. By comparison, reducing the fractional order from $\alpha = 1$ to 0.75 changed the amplitude by approximately 1.2%, while $\alpha = 0.5$ produced an amplitude close to the $\alpha = 1$ case. Under the conditions investigated, medium stiffness therefore had the clearer effect on radial amplitude.

The numerical convergence study showed systematic reduction in error with decreasing time step, with an observed convergence order of approximately 1.3–1.6. This is consistent with the fractional-memory discretisation limiting the overall accuracy of the coupled scheme despite the use of RK4 for the local state update.

The main limitation is that the fully coupled model has no direct external benchmark and that the fractional coefficient η_α is not independently calibrated from experimental fractional-order tissue data. The results should therefore be interpreted as a verified model formulation and a controlled numerical investigation rather than as an experimentally validated prediction of fractional-order tissue behaviour. A resonance-matched, phase-resolved study using experimentally calibrated rheological parameters is the most direct next step.

13 Acknowledgments

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A Supporting Derivations

The following derivations support the governing equations of Sections 5, 6, and 8. Steps already given in full in the main text are not repeated here; each entry focuses on the algebra bridging one main-text result to the next.

A.1 Radial Velocity Field

For an incompressible medium, conservation of volumetric flow rate Q across any spherical surface of radius r concentric with the bubble gives $Q = Av_r$, with $A = 4\pi r^2$ for a sphere, so

$$v_r(r, t) = \frac{Q}{4\pi r^2}.$$

At the bubble wall, $Q = 4\pi R^2 \dot{R}$ [24], giving the radial velocity field used in Section 5.1.2:

$$v_r(r, t) = \frac{R^2 \dot{R}}{r^2}.$$

A.2 Strain and Strain-Rate Fields

Differentiating v_r gives the radial and tangential strain rates, $\dot{\gamma}_{rr} = \partial v_r / \partial r = -2R^2 \dot{R} / r^3$ and $\dot{\gamma}_{\theta\theta} = v_r / r = R^2 \dot{R} / r^3$ (Section 5.1.3). Integrating $\dot{\gamma}_{rr}$ in time with the boundary condition $\gamma_{rr} = 0$ at $R = R_0$ gives

$$\gamma_{rr} = -\frac{2}{3r^3} (R^3 - R_0^3), \quad \gamma_{\theta\theta} = \frac{1}{3r^3} (R^3 - R_0^3).$$

These give the differences entering the momentum balance of Section 5.1.6:

$$\dot{\gamma}_{rr} - \dot{\gamma}_{\theta\theta} = -\frac{3R^2 \dot{R}}{r^3}, \quad \gamma_{rr} - \gamma_{\theta\theta} = -\frac{R^3 - R_0^3}{r^3}.$$

A.3 Linear Voigt Stress Contribution

Substituting the strain fields of Appendix A.2 into $\tau_{rr} = 2(G\gamma_{rr} + \mu\dot{\gamma}_{rr})$ gives

$$\tau_{rr} = -\frac{4G}{3r^3} (R^3 - R_0^3) - \frac{4\mu R^2 \dot{R}}{r^3}.$$

The pressure contribution follows from the stress integral $3 \int_R^\infty \tau_{rr} / r dr$. Since R and $\dot{R}$ do not depend on the integration variable r , and $\int_R^\infty r^{-4} dr = 1/(3R^3)$, this integral evaluates to

$$3 \int_R^\infty \frac{\tau_{rr}}{r} dr = -\frac{4G}{3} \left(1 - \frac{R_0^3}{R^3}\right) - 4\mu \frac{\dot{R}}{R}.$$

Substituting $x_1 = R/R_0$, $\tau = t/T$, and $x_2 = R_0/T \cdot \dot{R}$, and applying the pressure scaling $T^2/(\rho R_0^2)$, gives the dimensionless linear Voigt contribution of Eq. 8.

A.4 Fractional Voigt Stress Contribution

Replacing the integer-order viscous term $\mu\dot{\gamma}_{rr}$ with the Caputo fractional term $\eta_\alpha {}^C D_t^\alpha \gamma_{rr}$ [3] in $\tau_{rr} = 2(G\gamma_{rr} + \eta_\alpha {}^C D_t^\alpha \gamma_{rr})$ gives

$$\tau_{rr} = -\frac{4G}{3r^3} (R^3 - R_0^3) - \frac{4\eta_\alpha}{3r^3} {}^C D_t^\alpha (R^3 - R_0^3).$$

Using $R^3 - R_0^3 = R_0^3(x_1^3 - 1)$ and the fractional-derivative time-rescaling identity ${}^C D_t^\alpha = T^{-\alpha} {}^C D_\tau^\alpha$ [3], the stress integral of Appendix A.3 gives

$$3 \int_R^\infty \frac{\tau_{rr}}{r} dr = -\frac{4G}{3} \left(1 - \frac{1}{x_1^3}\right) - \frac{4\eta_\alpha}{3T^\alpha x_1^3} {}^C D_\tau^\alpha (x_1^3 - 1).$$

Applying the same pressure scaling as above gives the dimensionless fractional Voigt contribution of Eq. 9.

A.5 Nondimensional Keller–Miksis System

With $\tau = t/T$, $x_1 = R/R_0$, and $x_2 = dx_1/d\tau$, the physical radius, velocity, and acceleration are $R = R_0 x_1$, $\dot{R} = (R_0/T)x_2$, and $\ddot{R} = (R_0/T^2)dx_2/d\tau$. Substituting into the dimensional Keller–Miksis equation and dividing through by R_0^2/T^2 gives the nondimensional second-order form of Section 6.1. Introducing $x_2 = dx_1/d\tau$ as an auxiliary state variable reduces this to the first-order pair $dx_1/d\tau = x_2$ and the general expression for $dx_2/d\tau$ (Eq. 14), with the model-specific pressure difference Δp left unspecified until Section 6.3.

A.6 Yang Pressure Difference: Total Derivative and Final System

Since $\omega = 2\pi/T$ and $t = T\tau$, $\sin(\omega t) = \sin(2\pi\tau)$, giving the acoustic term used throughout. As Δp_Y depends on x_1 , x_2 , and τ (Eq. 20), its total derivative follows the multivariable chain rule (Eq. 21). Because Δp_Y depends on x_2 through the viscous term, $\partial \Delta p_Y / \partial x_2 \neq 0$, so the $dx_2/d\tau$ term arising in $d\Delta p_Y/d\tau$ must be collected onto the left-hand side of the Keller–Miksis

equation before it can be solved explicitly for $dx_2/d\tau$. Carrying this out gives the final system N_Y/D_Y of Eqs. 29–30, with D_Y carrying the additional viscous term $4\mu/(\rho c R_0)$ relative to the general form of Appendix A.5.

A.7 Marmottant Pressure Difference: Total Derivative

The radius-dependent surface tension $\sigma(x_1)$ (Eq. 37) requires the quotient-rule derivative

$$\frac{d}{dx_1} \left(\frac{\sigma(x_1)}{x_1} \right) = \frac{x_1 \sigma'(x_1) - \sigma(x_1)}{x_1^2},$$

with $\sigma'(x_1) = 0$ in the buckled and ruptured regimes and $\sigma'(x_1) = 2\chi R_0^2 x_1 / R_b^2$ in the elastic regime (Eq. 57). The same chain-rule and acceleration-collection procedure as Appendix A.6 applies to Δp_M , except that $\partial \Delta p_M / \partial x_2$ now includes both the liquid and shell viscosity terms, $-4\mu T / (\rho R_0^2 x_1) - 4\kappa_s T / (\rho R_0^3 x_1^2)$, giving the three-term denominator D_M of Eq. 35.

A.8 Fractional Model: Partial Derivatives and Caputo Memory Term

The partial derivatives of Δp_F (Eq. 39) with respect to x_1 , the fractional memory variable $\mathcal{M}_\alpha(\tau)$, and τ give the total derivative of Eq. 41. Since the fractional memory term does not introduce an instantaneous algebraic dependence on x_2 (unlike the linear viscous term of Appendix A.6), no $\partial \Delta p_F / \partial x_2$ term arises, and the acceleration-collection step of Appendices A.6–A.7 is unnecessary here.

The Caputo memory term (Section 5.2) in dimensionless time is, for $0 < \alpha < 1$ [3],

$${}^C D_\tau^\alpha f(\tau) = \frac{1}{\Gamma(1-\alpha)} \int_0^\tau \frac{\dot{f}(s)}{(\tau-s)^\alpha} ds.$$

With $f(\tau) = x_1^3(\tau) - 1$, so $\dot{f}(\tau) = 3x_1^2 x_2$, the memory term becomes

$$\mathcal{M}_\alpha(\tau) = \frac{1}{\Gamma(1-\alpha)} \int_0^\tau \frac{3x_1^2(s) x_2(s)}{(\tau-s)^\alpha} ds,$$

which is approximated numerically using the L1 discretisation described in Section 8.3.1.

B

Below is the code we have used within this project;

<https://github.com/S-EnS/Comparison-of-Bubble-Radius-Dynamics-in-Newtonian-and-Viscoelastic-Media>